

Luciano Maldonado

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EDUCATION

West Virginia University

Expected Graduation: May 2027

Bachelor of Science in Computer Science & minor in Applied Mathematics | GPA Overall: 3.7; Major GPA: 4.0

Extracurriculars/Organizations: Member of Artificial Intelligence Club at WVU; Member of ColorStack community; CodePath Member; LatinX in AI member.

SKILLS

Languages *Proficient* Java(2yrs) · Python(2yrs) *Intermediate* Javascript(2yrs) *Beginner* C++(1yrs) Typescript(1yrs) C(1yrs)

Software/OS AWS · Spring · Docker · Pytorch · TensorFlow · JupyterNotebook · iOS · Git · Bash · HuggingFace · Groq · Vercel · Anaconda · Prisma · Modal · LaTeX · Next.js · Neon · Drizzle · SQL · Linux · iOS · Windows

EXPERIENCE

Software Engineering Intern

June 2024 - August 2024

[Alpha Systems S.R.L](#)

La Paz, Bolivia

- Developed a **Task Management System** using Java, Spring Boot, and **PostgreSQL**
- Implement **CRUD operations and RESTful APIs**, resulting in a 5% improvement in task-tracking efficiency
- Consumed an endpoint to create an Excel report using **Java** and **Spring**, automating data extraction
- Develop **algorithm** for office desk assignments, remote work scheduling based on departmental needs using **Python**
- Optimizing resource allocation and optimizing remote and in-person work by 3% in the week of use

Software Engineering Resident

Sep 2024 - March 2025

Headstarter

Remote - New York, NY

- Built 14+ **machine learning, AI-engineering**, and full-stack projects in fast-paced software team environments
- Developed 5+ **neural networks** in Python, 11 apps in TypeScript on AWS/Vercel with dev and production environments
- Implemented **LLM-chaining**, hyperparameter tuning, **fine-tuning** on 10+ LLM models controlling for latency & accuracy,
- Coached by Google Machine Learning, Google Kubernetes, Two Sigma, Tesla, Figma, and Citadel Engineers

Undergraduate Research Assistant

Feb 2024 - Aug 2025

West Virginia University

Morgantown, WV

Dr Tehranipoor and Dr Karimian-Laboratory

- Research using **deep-learning techniques** to generate unique private keys from real ECGs recordings.
- Developed and benchmarked CNN + attention, ViT, and WaveNet models, **achieving up to 96%** intra-person consistency
- Led the **Post-Quantum Cryptography (PQC)** and Quantum Computing components during the SVCSI summer camp.
- Designed and evaluated **PrivDrift**, a multi turn benchmark for privacy leakage in LLMs across topic drift and persuasion settings
- Built a hybrid evaluation pipeline using pattern matching and LLM based judging to detect direct, partial, and paraphrased privacy leakage
- Coauthored a **PrivDrift** paper currently under review.

Machine Intelligence Lab under Dr Gyawali

Oct 2025 - Present

- Built an evaluation framework comparing **distilled vs non-distilled** LLMs on **reliability** and **sycophancy**.
- Ran automated **large-scale experiments** on an HPC to measure how distillation changes robustness, calibration, and harmful agreement.
- Created custom **metrics** and **data analysis** to pinpoint when distillation increases or reduces sycophancy.
- Extended the project into sycophancy evaluation with **Conformal risk Control (CCR)**, studying how uncertainty thresholds can mitigate sycophantic model responses.
- Coauthored a paper on LLM sycophancy evaluation with **CCR**, submitted to **MLHC'26** and currently under review.

Summer Undergraduate Research Intern (SURE program)

May 2025 - Aug 2025

West Virginia University

- Expanded three architectures for ECG key-generation into production-style **PyTorch** code with **WandB** tracking, optimizing training times and computation resources.
- Accepted and presented at the **SVCSI conference** for the **CNN+Multihead-attention** architecture, published on [IEEE](#).
- Built automated NIST Statistical Test Suite pipelines to validate 256-bit key randomness.

AI Researcher intern

Dec 2025 - Present

Algoverse

Remote

- Built with a team of 3, a **multi-agent RAG pipeline** that decomposes complex questions, retrieves evidence, and assembles grounded multi-hop answers.
- Integrated with **Google Gemini** for planning and reasoning with **FAISS** semantic search powered by local **SentenceTransformer** embeddings on **GPU** or **CPU**.

- Added **MCP (Model Context Protocol)** support for structured reasoning state management, alongside **end-to-end execution history** and **trajectory tracking**.
- Currently developing and evaluating a harness for a **RAG evaluation harness** using **EM** and **F1** metrics, and the work is in progress for publication.

Student Researcher Data Driven WV

Feb 2026 - Present

- Designed a **Python/pandas ETL pipeline** to unify 1.4M+ records across heterogeneous schemas.
- Implemented site-level deduplication logic with conflict-resolution rules and temporal consolidation.
- Automated **geospatial validation and quality-control checks** with reproducible artifacts and reports.
- Standardized categorical variables using **fuzzy matching and performed exploratory analysis** in notebooks.

AI PROJECTS

Agentic Support | Personal Project (~60 hours) - [GitHub Link](#) (Python, LangChain, Chroma, FastAPI, TypeScript) *April 2025 - June 2025*

- Orchestrated an agentic AI workflow with **LangGraph's framework** to handle **FAQs, refunds, and troubleshooting**.
- Developed a modular FastAPI backend with Uvicorn and Poetry for endpoint orchestration, using a **ReAct Agent** flow.
- Implemented semantic retrieval via **Chroma Vector Database (RAG)** and a model for the embeddings to store in Chroma.
- Enabled observability with **Langfuse** to track agent decisions, tool calls, and user interactions across workflows.

LLM Evaluation APP | Personal Project (~70 hours) - [GitHub Link](#) (Typescript, Next.js, Neon-DB, React, Tailwind CSS) *Jan 2025 - Feb 2025*

- Developed an evaluation system for **LLMs** using **cosine similarity** and scoring metrics.
- Implemented an **LLM-based judge** to assess model responses and provide automated evaluations.
- Designed a structured evaluation pipeline to compare LLM outputs against predefined benchmarks.

Codebase RAG APP | Personal Project (~60 hours) - [GitHub Link](#) (Python, Gradio, Groq, Huggingspaces, Pinecone) *Oct 2024 - Nov 2024*

- Built an **AI-powered web app** with **Gradio** for exploring and analyzing multi-language codebases.
- Developed a pipeline to clone and process GitHub repositories, leveraging **Tree-sitter** for **AST-based** parsing to generate embeddings and optimize code retrieval.
- Integrated **FastAPI**, **Pinecone**, and **Hugging Face** for scalable **vector storage** and **RAG workflows**.

PixelAlchemy | Personal Project (~50 hours) - [GitHub Link](#) (Python, Typescript, Modal, HuggingFace, Blob, React) *Nov 2024 - Dec 2024*

- Created an "Instagram-like" app generating images from text prompts, hosted on serverless **GPUs via Modal (Python)**.
- Implemented a **low-latency text-to-image** model for real-time generation, following best Generative AI practices.
- Developed a **TypeScript-based full-stack app** with smooth user interactions and database storage.